CSA 1718 – Artificial Intelligence

Name : P. Yaswanth

Reg No : 192524167

1. Program to Determine Whether a Particular Bird Can Fly or Not

Aim

To write and implement a Prolog program to determine whether a particular bird can fly or cannot fly.

Objective

- To understand facts and rules in Prolog.
- To represent bird characteristics using predicates.
- To query whether a particular bird can fly.

Theory

In Prolog, information can be represented using **facts** and **rules**. A fact directly states information, while a rule defines a relationship or condition.

For example:

- bird(parrot). represents that parrot is a bird.
- can_fly(parrot). represents that parrot can fly.

Algorithm

1. Define different birds as facts.
2. Define birds that can fly.
3. Define birds that cannot fly.
4. Query the required bird.
5. Prolog returns true if the bird can fly.

Program

```
bird(parrot).

bird(eagle).

bird(penguin).

bird(ostrich).
```

can_fly(parrot).

can_fly(eagle).

cannot_fly(penguin).

cannot_fly(ostrich).

Query

can_fly(parrot).

Output

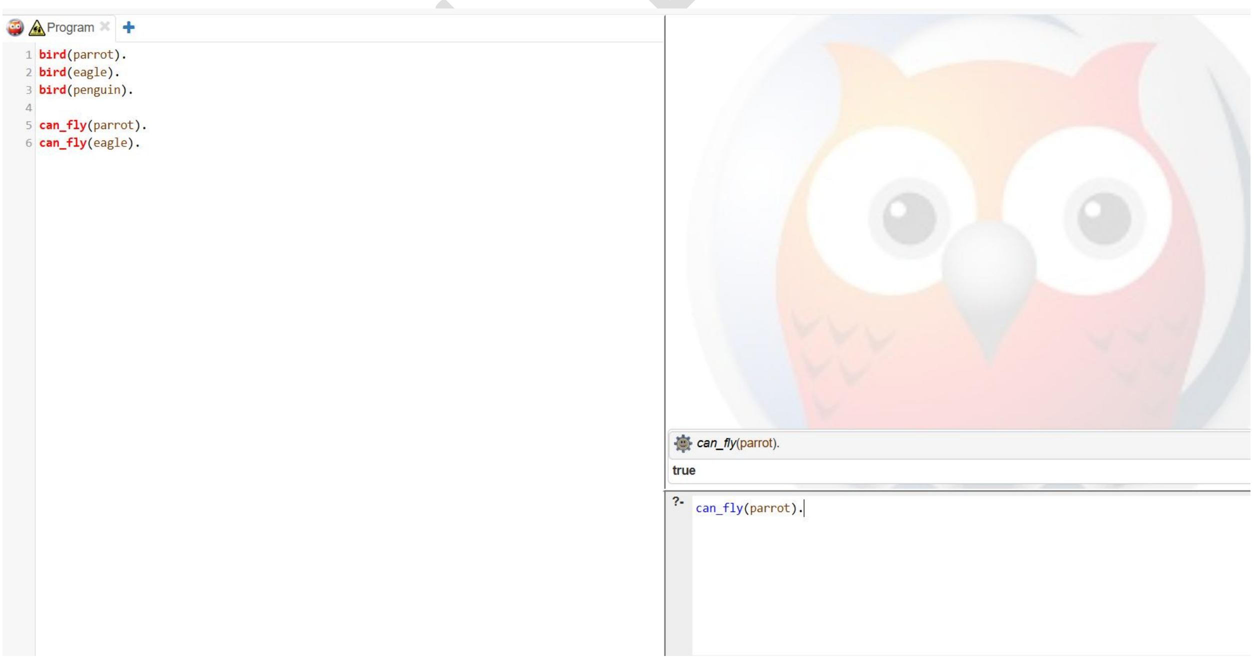
true.

Another query:

can_fly(penguin).

Output:

false.



Result

Thus, the Prolog program to determine whether a particular bird can fly or not was successfully implemented and executed.